25900528

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCMOS200 - IOT FOR ELECTRONICS
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

Part A

Multiple Choice Questions
Answer all questions , Each question carries **1** marks

1. **b)** To guide secure design, deployment, and operation of IoT systems
2. **d)** 6LoWPAN
3. **b)** #include <WiFi.h>
4. **b)** Devices are often deployed in remote areas
5. **b)** Need to work across borders and networks
6. **c)** Serial Monitor
7. **a)** Smart glasses
8. **c)** Security
9. **a)** 34–39
10. **b)** Taking action based on received commands

[10X1=10]

Part B

Short Answer Questions
Answer any **4** questions, Each question carries **5** marks

11. Challenges in IoT security & cloud infrastructure,
    Any **3 challenges**:
    Data privacy, authentication, scalability, data storage, cloud attacks, latency, firmware updates
    1+1+1+1+1
12. WoT in Smart Cities,
    Examples: Smart parking, waste management, traffic control, lighting; Explanation of how WoT enables device interoperability **2+3**

13. Gas sensor for LPG detection,
    MQ-2 sensor – detects LPG, CH_4; Working principle (resistance change); Features (low cost); Limitations (humidity sensitivity) **1+2+2**

14. Characteristics of IoT devices, Connectivity, scalability, energy efficiency, sensing, intelligence, interoperability; any 3 explained **1.5+1.5+2**

15. CoAP definition and suitability, Constrained Application Protocol; lightweight REST-based; UDP transport; low overhead; ideal for resource-constrained IoT nodes **2+3**

16. Output devices with ESP32, Examples: LED (status), Relay (control), Buzzer (alert), Display (data output); explain functions **1.5 +1.5+2**

[4X5=20]

Part C

Essay Questions
Answer any **2** questions, Each question carries **10** marks

17. Five key components of IoT, Components: Sensors, Connectivity, Data Processing, User Interface, Actuators; Functions and examples (e.g., temperature sensor, Wi-Fi, cloud, dashboard, motor) **2 MARKS EACH**

18. Objectives of OIC and support to IoT ecosystem, Objectives: Interoperability, security, openness, scalability, reliability; Explanation of how each promotes IoT ecosystem integration **5+5**

19. Program – Temperature & Light Read, Code using ESP32, DHT11/LM35 + LDR sensors, Serial.begin(), analogRead(), Serial.print(); correct syntax and comments **CODE-6, PGM-4**

20. IoTivity – Objectives & Features, Open-source framework; Features: Discovery, communication, data security, interoperability; Contribution to standardization (OIC-based) **5+5**

[10X2=20]

SCHEME PREPARED BY

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DEPT OG ELECTRONICS

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